

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 9: Class A Landing Performance & Factored Distance

Syllabus Source: CAE Oxford (pp. 317-360) | Pass Target: $\geq 90\%$ AviationExam

1. CLASS A LANDING REQUIREMENTS & SCREEN HEIGHT

Class A landing performance begins at a screen height of 50 ft above the runway threshold at target approach speed V_{ref} :

- Reference Landing Speed (V_{ref}): Target speed over threshold: V_{ref} MUST BE AT LEAST $1.23 \times V_{sr0}$ (stall speed in landing configuration).
- Landing Distance Available (LDA): The physical paved runway declared available from threshold to end of roll.
- The 60% Rule (Dry Runway Factor $\times 1.67$): Under EASA CAT.POL.A.230, a turbojet aircraft must be capable of landing from 50 ft to a complete stop within 60% OF THE LDA! Therefore:
Required Factored Landing Distance = Actual Measured Distance / 0.60 = Actual Distance $\times 1.67$.
- Wet Runway Factor ($\times 1.15$ of Dry = $\times 1.92$ Total!): If the runway is forecast to be wet or slippery at ETA, the required landing distance is 115% OF THE FACTORED DRY DISTANCE:
Required Wet Landing Distance = Actual Distance $\times 1.67 \times 1.15$ = Actual Distance $\times 1.92$!

2. REVERSE THRUST REGULATORY CREDIT

Under CS-25 certification for dispatch purposes, NO CREDIT is permitted for thrust reversers on dry or wet runways! The aircraft must be capable of stopping using wheel brakes and aerodynamic spoilers alone. (Reverse thrust credit is only considered in non-normal in-flight assessments or contaminated runway models).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Class A Jet Landing Factored Dis

Rules for Commercial Jet Dispatch: Step 1: Measured AFM landing distance from 50 ft threshold to full stop. Step 2: Dry Runway Factor = $\times 1.67$ (Must stop within 60% of LDA: $1 / 0.60 = 1.67$). Step 3: Wet Runway Factor = Dry Factored Distance $\times 1.15$. Step 4: Crucial Exam Rule: Reverse thrust credit is FORBIDDEN in dry dispatch landing calculations! Spoilers and wheel brakes only.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Dispatch Landing Distance Check Dry vs Wet

SCENARIO (Commercial Airline Dispatch Drill): A Boeing 737 is dispatching to a destination with a Landing Distance Available (LDA) of 2,200 meters: - Flight Manual Unfactored Landing Distance from 50 ft = 1,200 meters - Destination forecast: Rain showers at ETA (Runway WET!) QUESTION: What is the Required Dry Landing Distance? What is the Required Wet Landing Distance? Can the aircraft legally dispatch? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Required DRY Landing Distance (The 60% Rule = $\times 1.67$): - Required Dry Distance = Actual Distance / 0.60 = 1,200 m / 0.60 = 2,000 meters. - (On a dry runway, 2,000 m $\leq$ 2,200 m LDA $\rightarrow$ Legal for dry runway). Step 2: Calculate Required WET Landing Distance (Apply 115% factor to Dry): - Required Wet Distance = Required Dry Distance $\times 1.15$ - Required Wet Distance = 2,000 m $\times 1.15$ = 2,300 meters! - (Notice: 1,200 m $\times 1.92$ = 2,304 meters). Step 3: Compare Required Wet Distance with Runway LDA: - Required Wet Distance = 2,300 meters - Available LDA = 2,200 meters - Verdict: 2,300 m $>$ 2,200 m $\rightarrow$ EXCEEDS AVAILABLE RUNWAY BY 100 METERS! FINAL ANSWER: The aircraft CANNOT legally dispatch! Landing mass must be reduced, or an alternate destination selected.

[!] EXAM TRAP (AVIATIONEXAM) - The 60% Rule vs 1.67 Factor Distinction Trap

- Stating 'Actual distance must not exceed 60% of LDA' is mathematically identical to stating 'Required LDA must be at least Actual distance multiplied by 1.67 ($1 / 0.60 = 1.667$)'! - For Wet runways, multiply by 1.92 total! Always check whether the exam question asks for DRY or WET!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA CS-25, what is the minimum approach speed (V_{ref}) over the runway threshold for a commercial aircraft?

- [A]: $V_{ref} \geq 1.23 \times V_{sr0}$ (stall speed in landing configuration)
- [B]: $V_{ref} = V_{sr0}$
- [C]: $V_{ref} = 1.10 \times V_{mca}$
- [D]: Exactly 140 kt

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. CS-25.125 requires that V_{ref} be at least 23% above the reference stall speed in the landing configuration: $V_{ref} \geq 1.23 V_{sr0}$.

Q2: For a commercial turbojet landing on a DRY runway, what proportion of the Landing Distance Available (LDA) must be used?

- [A]: Exactly 50%
- [B]: Not more than 60% of the LDA (factoring distance by 1.67)
- [C]: 70%
- [D]: 100%

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CAT.POL.A.230, the landing distance from 50 ft must not exceed 60% of LDA, ensuring a 40% runway safety buffer.

Q3: When calculating dispatch landing performance on a WET runway under CS-25, what additional factor is applied to the dry landing distance?

- [A]: 10%
- [B]: An additional 15% (giving a total factor of 1.92 times actual distance)

- [C]: 30%
- [D]: Zero factor

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A wet runway requires a 15% increase over the dry factored distance (Factored Dry x 1.15), reflecting reduced wet tire friction.